

Final CCNA Enterprise Network Project

Enterprise Office Physical Network Infrastructure Design

Project Overview

This project is a professional enterprise-style network topology designed using Cisco Packet Tracer as part of a CCNA learning journey. The focus of this project is the topic: **Interfaces and Cables**.

Enterprise Objective

Design a scalable enterprise office network that connects multiple departments through access switches, a centralized core switch, routing infrastructure, firewall protection, and internet connectivity.

Real Business Problem

A medium-sized company requires a reliable and organized physical network infrastructure to support employee communication, internet access, shared resources, printers, servers, and IP phones while preparing for future scalability.

Department Structure

Administration Department

- PCs
- Printer
- IP Phone

Engineering Department

- PCs
- Laptops
- Server

Management Department

- PCs
- Printer
- Server
- IP Phone

Enterprise Topology

```
Administration -> SW-ADMIN
Engineering    -> SW-ENG
Management    -> SW-MGMT
```

```
SW-ADMIN SW-ENG    -> CORE-SW -> EDGE-RTR -> ASA Firewall -> Internet
SW-MGMT   /
```

OSI Layer Concepts

- Switches operate at Layer 2 using MAC addresses
- Routers operate at Layer 3 using IP addresses
- Firewalls provide Layer 3/4+ security filtering

Cable Selection Rules

- PC -> Switch = Straight-through cable
- Switch -> Router = Straight-through cable
- Switch <-> Switch = Cross-over cable
- Router -> Firewall = Straight-through cable
- Firewall -> Internet = Straight-through cable

Cisco Interfaces

- Fa0/1 = FastEthernet 0/1
- Gig0/0 = GigabitEthernet 0/0
- Gig0/1 = GigabitEthernet 0/1
- Console Interface

Skills Demonstrated

- Enterprise topology design
- Interface identification
- Cable selection
- Layer 2 and Layer 3 concepts
- Core switch architecture
- Basic troubleshooting awareness

Future Improvements

- VLANs
- Inter-VLAN Routing
- DHCP
- ACLs
- OSPF
- Network Security

Router Interface Activation Commands

```
enable
configure terminal
interface gigabitEthernet 0/0
no shutdown
exit
interface gigabitEthernet 0/1
no shutdown
show ip interface brief
```

Devices Used

Device	Quantity	Purpose
PCs	6	Employee Workstations
Laptops	3	Mobile Devices
Printers	2	Shared Printing
IP Phones	2	Voice Communication
Servers	2	Internal Services
Switches	4	Access/Core Switching
Router	1	Routing
Firewall	1	Security Filtering
Cloud	1	Internet/WAN